

Q1.

Sketch the graph given by the polar equation

$$r = \frac{a}{\cos \theta}$$

where a is a positive constant.



(Total 2 marks)

Q2.

A curve has polar equation $r = 3 + 2 \cos \theta$, where $0 \leq \theta \leq 2\pi$

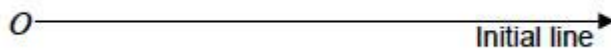
- (a) (i) State the maximum and minimum values of r .

(2)

- (ii) Sketch the curve.

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(2)

- (b) The curve $r = 3 + 2 \cos \theta$ intersects the curve with polar equation $r = 8 \cos^2 \theta$, where $0 \leq \theta \leq 2\pi$

Find all of the points of intersection of the two curves in the form $[r, \theta]$.

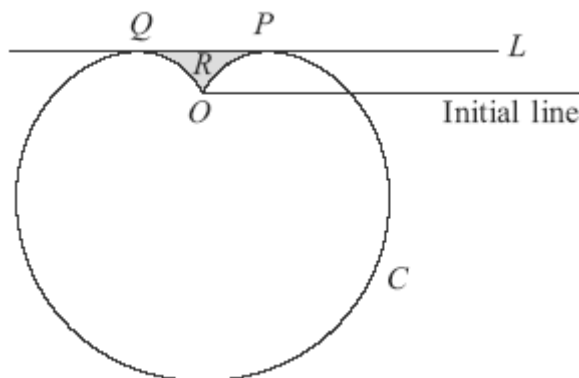
You should leave your answers in exact form where possible.

(6)

(Total 10 marks)

Q3.

The diagram shows a sketch of a curve C and a line L , which is parallel to the initial line and touches the curve at the points P and Q .



The polar equation of the curve C is

$$r = 4(1 - \sin \theta), \quad 0 \leq \theta < 2\pi$$

and the polar equation of the line L is

$$r \sin \theta = 1$$

- (a) Show that the polar coordinates of P are $\left(2, \frac{\pi}{6}\right)$ and find the polar coordinates of Q .

(5)

- (b) Find the area of the shaded region R bounded by the line L and the curve C . Give your answer in the form $m\sqrt{3} + n\pi$, where m and n are integers.

(11)

(Total 16 marks)

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Q4.

The polar equation of a curve C_1 is

$$r = 2(\cos \theta - \sin \theta), \quad 0 \leq \theta \leq 2\pi$$

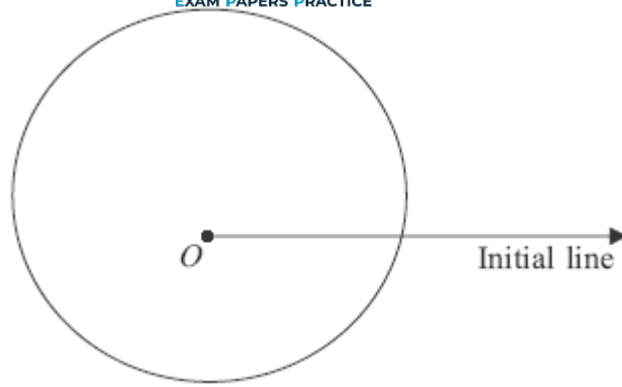
- (a) (i) Find the cartesian equation of C_1 .
- (ii) Deduce that C_1 is a circle and find its radius and the cartesian coordinates of its centre.

(4)

(3)

- (b) The diagram shows the curve C_2 with polar equation

$$r = 4 + \sin \theta, \quad 0 \leq \theta \leq 2\pi$$



- (i) Find the area of the region that is bounded by C_2 . (6)
 - (ii) Prove that the curves C_1 and C_2 do not intersect. (4)
 - (iii) Find the area of the region that is outside C_1 but inside C_2 . (2)
- (Total 19 marks)**

Q5.

The diagram shows the curve C_1 with polar equation

$$r = 1 + 8e^{-\frac{\theta}{\pi}}, \quad 0 \leq \theta \leq 2\pi$$



- (a) Find, in terms of π and e , the area of the shaded region bounded by C_1 and the initial line. (5)
- (b) The polar equation of a curve C_2 is

$$r = e^{\frac{\theta}{\pi}}, \quad 0 \leq \theta \leq 2\pi$$

Sketch the curve C_2 and state the polar coordinates of the end-points of this curve. (4)

- (c) The curves C_1 and C_2 intersect at the point P . Find the polar coordinates of P . (5)
- (Total 14 marks)**

Q6.

A curve C has polar equation

$$r^2 \sin 2\theta = 8$$

- (a) Find the cartesian equation of C in the form $y = f(x)$. (3)
- (b) Sketch the curve C . (1)
- (c) The line with polar equation $r = 2 \sec \theta$ intersects C at the point A . Find the polar coordinates of A . (4)

(Total 8 marks)

Q7.

The polar equation of a curve C is

$$r = 5 + 2 \cos \theta, \quad -\pi \leq \theta \leq \pi$$

- (a) Verify that the points A and B , with **polar coordinates** $(7, 0)$ and $(3, \pi)$ respectively, lie on the curve C . (2)
- (b) Sketch the curve C . (2)
- (c) Find the area of the region bounded by the curve C . (6)
- (d) The point P is the point on the curve C for which $\theta = \alpha$, where $0 < \alpha \leq \frac{\pi}{2}$. The point Q lies on the curve such that POQ is a straight line, where the point O is the pole. Find, in terms of α , the area of triangle OQB . (4)

(Total 14 marks)